

Databases – Exercises Transactions

Vrije Universiteit Amsterdam

1. Consider the following schedule:

T1	W(X)
T2	R(Y) R(X)
T3	W(Z) W(Y)

Show that the following schedule is realisable with two-phase locking. For this purpose, insert all necessary lock and unlock actions.

2. Let us revisit the schedule from the previous task:

T1	<i>start W(X) commit</i>
T2	<i>start R(Y) R(X) commit</i>
T3	<i>start W(Z) W(Y) commit</i>

The schedule now explicitly indicates the start and end of each transaction.

- (a) Can this schedule be achieved using strict two phase locking?
Motivate your answer.
- (b) Can this schedule be achieved using preclaiming two phase locking?
Motivate your answer.

3. Consider the following schedule:

T1	<i>start R(X) W(Y) commit</i>
T2	<i>start R(X) W(Y) commit</i>

- (a) Can this schedule be achieved using strict two phase locking?
Motivate your answer.
- (b) Can this schedule be achieved using preclaiming two phase locking?
Motivate your answer.

4. Consider the following schedule:

T1	<i>start W(X) R(X) commit</i>
T2	<i>start R(X) commit</i>

- (a) Can this schedule be achieved using strict two phase locking?
Motivate your answer.
- (b) Can this schedule be achieved using preclaiming two phase locking?
Motivate your answer.

5. Consider the following schedule:

T1	R(Z)	W(Y)	
T2	W(X)		R(Y)
T3			R(Z)
T4	W(Z)		R(X)
T5		W(Z)	
T6		R(X)	

Assume that transaction 4 is aborted/rolled back. Is this rollback cascading? What other transactions must be rolled back?

6. Consider the following schedule:

T1	R(X)	W(Y)	W(Z) ?
T2		W(X)	R(Y) ?
T3		R(X)	R(A) ?
T4			R(Z) W(A) ?

Assume that the database management system uses backward oriented optimistic concurrency control. The question marks indicate the attempt of the transactions to commit. Determine the read sets and write sets for each transaction. Explain what happens with these transactions when they try to commit?